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The NTR/Prodrug Revolution: Tools for Controlling Cell Loss and Regeneration

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eLife Assessment

This Review Article synthesizes the development, applications, and recent technical advances of the nitroreductase/prodrug system, highlighting how it enables precise spatiotemporal cell ablation and experimental platforms for studying regenerative mechanisms and screening for pro-regenerative or protective compounds. Together, the article provides a conceptual and practical overview that will help researchers adopt and further develop this versatile approach in regenerative biology. It will be of interest to researchers studying regeneration, disease modelling, and targeted cell ablation, particularly those working with zebrafish and other genetic model systems.

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Abstract

Here, we review the history, advancements, and broad utility of the NTR/prodrug system, and suggest future strategies for developing versatile ablation models. As a chemogenetic tool, the nitroreductase (NTR)/prodrug system enables precise spatiotemporal control over cell ablation. The technology leverages bacterial nitroreductase enzymes (e.g., *nfsB*) to convert inert prodrugs into cytotoxic agents, thereby allowing researchers to induce targeted cell death. Following its landmark application in zebrafish with metronidazole (MTZ) in 2007, the system's utility has expanded to other essential model organisms, including *Drosophila, Nematostella, Xenopus*, medaka, and rodents, facilitating detailed studies of tissue damage and regeneration. This review highlights how the NTR system has been deployed to model a spectrum of human diseases, including Parkinson's disease, retinal degeneration, demyelinating disorders, and kidney disease. These models provide valuable platforms to study pathogenesis *in vivo*. Furthermore, the precise and controllable nature of NTR ablation makes it an ideal tool for high-throughput chemical and genetic screens aimed at discovering pro-regenerative and protective compounds. The development of NTR2.0, an enzyme variant with over 100-fold greater activity, along with more potent prodrugs such as ronidazole (RNZ), has dramatically broadened experimental possibilities. These improvements permit chronic ablation and long-term disease modeling at well-tolerated drug concentrations. Here we present some key considerations including transgenic design for optimal cell-type specificity, calibrating expression levels for desired ablation kinetics, and suitable controls to allow interpretation. These best practices will allow the researcher to develop a precise, reproducible, and versatile platform for either modeling human disease or dissecting regenerative mechanisms.

Introduction

Why regeneration?

We often regard ourselves as having limited regenerative capacity, yet numerous human tissues exhibit substantial renewal: the liver restores lost mass, the skin undergoes continuous stem cell–driven turnover, skeletal muscle repairs through satellite cells, bone heals via remodeling, blood is replenished by hematopoietic stem cells, the intestinal epithelium renews within days, and the endometrium regenerates cyclically. However, the regenerative capacity of human tissues is modest when contrasted with that of many other species, in which entire organs or appendages can be restored following injury [1,2]. This capacity is profound in many invertebrates; for instance, hydra and planarians can regenerate complete organisms from minute fragments [3].

Understanding these enviable examples provides a framework for uncovering the cellular and molecular mechanisms that organisms employ to repair or replace damaged tissues, with the ultimate goal of developing new strategies to combat human injuries, degenerative diseases, and age-related decline [4,5]. An essential requirement for this research is the ability to induce controlled, reproducible injury to study the subsequent repair processes.

The laboratory mouse *(Mus musculus)* is a critical vertebrate model for human biology and has been essential for developing inducible, cell-type-specific genetic tools in mammals. However, its intrinsic regenerative capacity is largely confined to a neonatal period, with abilities such as heart and digit tip regeneration declining sharply in adulthood [6]. The spiny mouse *(Acomys)* exhibits robust adult regeneration of skin, muscle, and ear tissue [7], but as an emerging model, it lacks the genetic toolkit for the systematic dissection of regenerative mechanisms.

For developing precise ablation tools, a genetically tractable and cost-effective vertebrate with robust adult regeneration is desirable. The zebrafish (*Danio rerio*) is uniquely suited for this role. Its external development, transparency, and high fecundity facilitate direct *in vivo* visualization and high-throughput screening [8]. Critically, zebrafish exhibit widespread regenerative abilities in adulthood, fully restoring complex organs including the heart [9,10], retina [11,12], spinal cord [13,14], and fins [15,16]. These attributes make zebrafish a practical system for dissecting the mechanisms of vertebrate regeneration. [17]

Methods of ablation

Researchers study regeneration by damaging tissue in different ways. The regenerative process that follows depends heavily on how the injury was created. The main types of injury model used in zebrafish can be classed as follows:

1) Physical Injury:

Direct tissue damage via surgical intervention that have been used successfully to study adult regeneration, including fin amputation, [18] ventricular resection, [9] brain lesioning, [19] scale removal [20,21] and transection of both tendons [22] and ligaments. [23] Severing of the spinal cord predictably leads to paralysis in zebrafish, but as testament to their regenerative capacity, full mobility is returned by 8 weeks. [13,24,25]

Several **no**n-surgical physical methods can be used to induce tissue damage in animal models, with their suitability largely determined by the subject's size and developmental stage. In adult models, approaches include cryoinjury to simulate myocardial infarction [26,27], **i**ntense light exposure to trigger photoreceptor degeneration in retinal regeneration studies [28], and acoustic trauma to model inner-ear damage and hearing loss recovery [29,30]. The acoustic method has also been applied to larval zebrafish to target lateral line hair cells [31]. Larval and embryonic zebrafish, owing to their small size and optical transparency, permit highly precise laser ablation of single neurons [32-34] or even individual cardiomyocytes [35]. Additionally, thermal injury has been employed in larvae to model burns and skin regeneration, revealing rapid inflammatory cell recruitment [36] and keratinocyte migration, but impaired sensory axon regeneration compared to mechanical injury [37].

2) Cell-Specific Toxins:

While physical injury has provided key insights into regenerative biology, these approaches are often constrained by limited precision and cell-type specificity. Pharmacological methods offer a complementary strategy, using small molecules to induce targeted, dose-controlled damage. For instance, selective toxins are used to ablate specific cell populations:

- Aminoglycoside antibiotics ablate sensory hair cells [38-40].
- Streptozotocin (STZ) eliminates pancreatic β-cells [41]
- Ouabain targets retinal neurons [42,43].
- Neurotoxins such as l-methyl-4-phenyl-l,2,3,6-tetrahydropyridine (MPTP) and 6-hydroxydopamine (6-OHDA) kill dopaminergic neurons, modeling aspects of Parkinson's disease [44].

However, these drugs are also hampered by a lack of true cellular specificity, often producing significant off-target effects that complicate experimental interpretation. Agents such as MPTP have the additional caveat of requiring strict handling procedures due to their high toxicity. To achieve higher precision and reduce off-target effects as well as expand the kinds of cell that can be ablated, the field has increasingly turned to genetic strategies. These methods allow damage to be targeted with exquisite specificity to predefined cell types.

3) Optogenetic cell ablation:

Due to the transparent nature of young zebrafish, this model system is very amenable to optogenetic techniques that allow for precise, non-invasive control of cellular signaling, neuronal circuit activity, and targeted cell ablation (see reviews [45,46]. KillerRed is a genetically encoded photosensitizer that, when illuminated with green or yellow light, produces reactive oxygen species (ROS) to kill nearby cells. [47,48] By driving its expression with cellspecific promoters, researchers can eliminate targeted cell types in living embryos or larvae with precise spatiotemporal control. [47-49] A key advantage of this technique is the speed of ablation, as transgene-expressing cells can be eliminated within hours of light exposure. This approach is particularly useful for modeling diseases in which ROS-mediated cell death plays a central role, such as neurodegeneration, [50] and cardiomyopathies, [51] but has not been applied in conventional regeneration studies.

4) Chemogenetic, cell-specific ablation:

Chemogenetic ablation strategies use a transgene to drive cell-specific expression of an exogenous protein, rendering target cells susceptible to a specific chemical agent. Four such systems established in zebrafish are:

i) Human Diphtheria Toxin Receptor and Diphtheria toxin:

The human pathogen *Corynebacterium diphtheriae* produces diphtheria toxin (DT). [52] This toxin binds to a specific human cell surface receptor called the diphtheria toxin receptor (hDTR) to enter and kill cells. [53] Most animals are resistant to DT because the toxin does not bind their endogenous receptors. A powerful experimental technique involves genetically engineering model organism to express the hDTR in a specific cell type. [54] When diphtheria toxin is injected into these animals, it only enters and ablates the cells expressing the hDTR, leaving all other cells unharmed.

Although widely used in mouse studies, its application in zebrafish is less common. [55,56] An exciting study by Jimenez et al. (2021) generated a transgenic zebrafish line in which the human diphtheria toxin receptor (hDTR) was expressed under the control of the hair-cell specific *myo6b* promoter. Upon diphtheria toxin administration, hair cells in the utricles, saccules, and lateral line were ablated in a dose-dependent manner in both larvae and adult zebrafish. Of note, complete regeneration of these sensory hair cells occurred within days, underscoring the value of this model for investigating targeted cell ablation and the mechanisms of hair cell regeneration across developmental stages. [55] Some studies have used transgenics to express subunit A of diphtheria toxin (DTA) in target cells, [57,58] but constitutive DTA expression causes cell loss without temporal control, precluding regeneration studies. Others have incorporated Cre/lox inducible systems to temporally control DTA expression and allow controlled ablation followed by analysis of recovery, [59] although leakiness, recombination efficiency, and transgene mosaicism can limit precision. [60]

ii) Inducible caspase systems:

Caspase cascade in apoptosis begins with upstream initiator caspases (e.g., caspase 8) activated by dimerization, which then activates effector caspases (e.g., caspase 3) to execute cellular dismantling. [61-63] In zebrafish, researchers have exploited induction of caspase-8 dimerization to achieve temporal and spatial control of cell-specific ablation in two main ways: 1) activated by the FK1012 chemical inducer of dimerization; [56,64 and, 2) expression of a fusion between caspase and the modified estrogen receptor ligand-binding domain (ERT2), whose activity is induced by binding of tamoxifen. [65-67] Compared to FK1012, tamoxifen pharmacokinetics are better characterized in zebrafish, and a convincing study showing lifelong regeneration of cerebellar Purkinje cells in zebrafish strongly supports the use of the ER-T2/tamoxifen inducible ablation approach for probing cell loss and recovery. [65]

iii) HSV-TK:

The herpes simplex virus thymidine kinase (HSV-tk) has been applied in zebrafish for conditional cell ablation. Transgenic expression of this “suicide” gene in a defined cell population converts the antiviral prodrug ganciclovir into toxic nucleotides that are lethal to proliferating cells, but this dependence on cell division has limited the utility of the approach. [68]

iv) Nitroreductase (NTR):

Bacterial nitroreductase genes (nfsB) are another class of “suicide” genes. These genes encode nitroreductases (NTR), enzymes that convert nitro-containing prodrugs (e.g., nitroimidazoles and nitrofurans) into cytotoxic compounds. As a result, any transgenic zebrafish cell expressing NTR becomes vulnerable to treatment with prodrugs such as metronidazole (MTZ). This approach to cell-specific ablation has become a widely adopted tool in the scientific community. Although first applied as an ablation tool in zebrafish, NTR/MTZ-mediated ablation has also been adapted for other model organisms such as, NTR/MTZ-mediated ablation has also been adapted for other model organisms; such as: *drosophila,* [69] *nematostella,* [70] *xenopus,* [30,71] medaka, [72] rats [73] and mice. [74] To date in zebrafish, the NTR/MTZ system has been used in a wide variety of different tissue to both study regeneration and model human pathology (Fig. 1 and Sup. Table 1). However, historically, interest in NTR was first driven by its potential as a “suicide gene” for cancer therapy.

Development of Nitroreductase as a Suicide Gene

Bacterial nitroreductases (NTRs) from *E. coli* (nfsA and nfsB) were first characterized in the 1970s–80s for their role in reducing nitroaromatics, [75-77] a function later harnessed in the 1990s for Directed Enzyme Prodrug Therapy (DEPT). [78-80] DEPT strategies use viral or antibody-based systems to deliver a ‘suicide gene’ or its enzyme product to tumors (e.g. NTR). [81] Once localized, the enzyme converts an administered prodrug into a cytotoxic agent, selectively killing the cancer cells.

The prodrug CB1954 is harmless to human cells but becomes a powerful, DNA-damaging toxin after activation by NTR. [82,83] Even though the NTR/CB1954 approach has been tested in clinical trials, [84,85] its effectiveness in mitigating cancers is limited due to low NTR activity and slow prodrug metabolism. Despite these therapeutic shortcomings, this work established NTR/CB1954 as a potent conditional cell-killing strategy. However, a key feature of the system, the “bystander effect,” where cytotoxic metabolites diffuse to kill neighboring cells, poses a significant problem for its application in regeneration studies. This effect complicates the interpretation of cell-specific ablation and subsequent regenerative outcomes. To overcome these limitations, researchers turned to alternative prodrugs. Metronidazole (MTZ), a nitroimidazole antibiotic, emerged as a particularly effective option because it is non-toxic to eukaryotic cells until reduced by bacterial NTR. Unlike CB1954, the activated metabolites of MTZ are short-lived and largely confined within

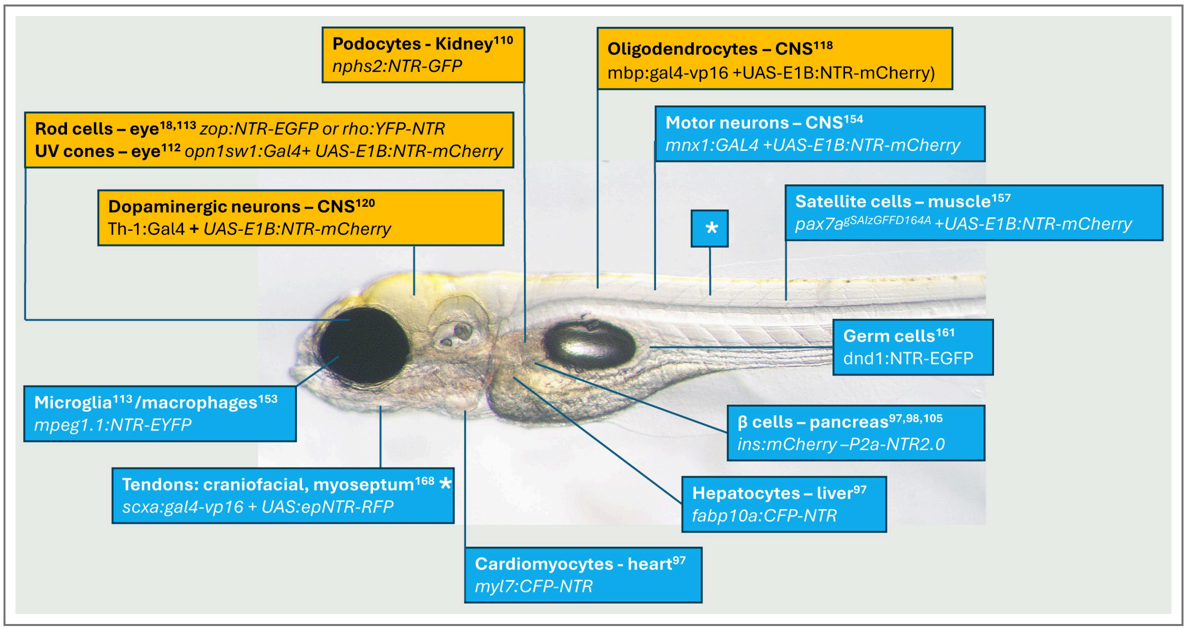

Figure 1. **Commonly targeted cell types for ablation studies in zebrafish.**

See S.Table 1 for more complete list. Image of a 5 days post-fertilization (dpf) *Casper* zebrafish larval59 with approximate position of cell type ablated. Name of cell, and transgene provided along with reference. Orange highlights indicate transgenic models used to study human pathologies.

the target cell, minimizing bystander effects [86-88]. This property made the NTR/MTZ system especially well-suited for regeneration and developmental studies, where precise, cell-specific ablation is essential.

NTR/MTZ in regenerative studies

The NTR/MTZ system for cell-specific ablation was first described in two 2007 studies. Pisharath et al. placed the *E. coli* nfsB gene (NTR) under the zebrafish insulin promoter to express an NTR–mCherry fusion in pancreatic β cells; treatment with 10 mM MTZ produced complete β-cell loss without affecting neighboring *α* cells or exocrine tissue. [89] Curado et al. used cell-specific promoters to express CFP-NTR in cardiomyocytes, hepatocytes, and β cells, also showing that MTZ induced targeted cell death with no detectable bystander effects. [90] In each case, tissues regenerated after MTZ withdrawal, establishing the NTR/MTZ system as a versatile, specific, inducible, and reversible tool for regeneration studies, [87,91] with NTR functional whether fused to either the N- or C-terminus of a fluorescent reporter.

Once targeted ablation is established, the next step is to identify the cellular source of regeneration. This is best accomplished by combining ablation with lineage tracing. For example, following the ablation of β-cells or hepatocytes, [89,90] researchers used this approach to trace the origins of new regenerated cells. In both the pancreas and liver, Notch-responsive ductal epithelial cells were identified as facultative progenitors responsible for repopulating the tissue. [92-95] Pancreatic centroacinar cells (CACs) and liver biliary epithelial cells (BECs/cholangiocytes) were shown to delaminate from ducts, dedifferentiate into precursors, proliferate, and then redifferentiate to replenish lost β cells or hepatocytes as needed. Combining single-cell RNA sequencing (scRNA-seq) of pancreatic ducts and hepatic ducts after β-cell/hepatocyte ablation has been used to map molecular mechanisms and identify intermediate progenitor states as new β-cells/hepatocytes are formed. [96,97] This integrated paradigm of targeted ablation, lineage tracing, and scRNA-seq systematically maps the mechanisms of regeneration, providing a crucial roadmap for developing future regenerative therapies.

NTR/MTZ - Dependent Ablation in Modeling Human Disease

The NTR/MTZ ablation system also lends itself well to modeling human diseases that are characterized by the specific and progressive loss of distinct cell populations. The core strengths of this chemogenetic approach include cell-type specificity, and temporal control which allows it to be a toolkit for recapitulating pathological events *in vivo*. This allows for the real-time dissection of disease initiation, progression, and complex cellular responses to injury. A diverse array of human pathologies has been modeled using NTR/MTZ in zebrafish, including chronic hyperglycemia (a symptom of diabetes), [98] acute liver damage, [92] and cardiac injury. [90,99,100] Here, we analyze selected models in greater detail, focusing on kidney disease, retinal degeneration, demyelinating disorders, and neurodegeneration. (Fig. 1 - orange boxes).

Kidney glomerular disease

Glomerular diseases stem from a common problem: progressive podocyte loss or dysfunction. These cells are crucial for maintaining the kidney's filtration barrier and their damage leads to proteinuria, the leakage of abnormal amounts of protein into the urine. [101] To model this pathology, transgenic zebrafish (e.g., *nphs2:NTR-GFP)* were used, where the *nphs2* promoter drives NTR expression to allow precise, inducible podocyte ablation. [102,103] Administering MTZ triggers rapid podocyte apoptosis, resulting in classic features of human glomerular injury causing disrupted filtration barrier with proteinuria and edema. By mirroring these essential aspects, this model provides a direct and relevant system for studying the progression of human podocytopathies. Furthermore, the zebrafish pronephros enables live imaging of podocyte injury and subsequent regeneration. [102] After MTZ withdrawal, podocyte repopulation occurs through residual cells and local progenitors. [103] This makes the model an ideal platform for investigating the cellular and molecular basis of podocyte repair, with direct relevance for discovering therapeutic pathways in human kidney disease.

Retinal degeneration

Given the high conservation of eye structure between zebrafish and humans, the NTR/MTZ system provides an excellent platform for modeling inherited retinal degenerations such as retinitis pigmentosa and cone dystrophies, conditions in which progressive photoreceptor loss leads to vision decline. [104] By driving NTR expression under photoreceptor-specific promoters, distinct photoreceptor subtypes can be selectively ablated. For example, expression under the rhodopsin (rho) promoter enables targeted elimination of rod photoreceptors, providing a robust zebrafish model of retinitis pigmentosa. [12] Similarly, the use of cone opsin promoters such as *opn1sw1* permits ablation of defined cone populations to study cone dystrophies. [105] This targeted ablation triggers apoptotic photoreceptor loss while sparing neighboring retinal cells. [106] A major advantage of the zebrafish system is its capacity for spontaneous retinal regeneration, driven by the dedifferentiation and proliferation of Müller glia that give rise to new photoreceptors. [107,108] The NTR/MTZ paradigm allows for precise initiation and synchronization of this regenerative process, enabling realtime dissection of the cellular and molecular programs underlying photoreceptor replacement and the contributions of innate immune signaling to retinal repair. [109]

Demyelinating disorders

Conventional autoimmune models of Multiple Sclerosis (MS), including Experimental Autoimmune Encephalomyelitis (EAE), often display substantial variability in the timing and severity of disease onset, alongside a highly complex and multifactorial immunopathology. [110] This inherent heterogeneity makes it difficult to disentangle the individual cellular and molecular events that specifically contribute to successful remyelination, thereby limiting the ability to clearly define the mechanisms required for effective tissue repair. To overcome these hurdles, NTR-MTZ based models have been utilized to ablate oligodendrocytes and their progenitors. Tg(*mbp:gal4-vpl6*); Tg(*UAS-E1B:NTR-mCherry*) fish express NTR specifically in mature oligodendrocytes that myelinate CNS axons. Exposure to MTZ in these fish caused rapid and synchronized demyelination within 48 hours, characterized by the retraction of myelin sheaths and oligodendrocyte cell death. [111] Furthermore, subsequent regeneration resulted in myelin sheaths that restored normal length and thickness correlated to axon caliber. [112] This mechanistic parallel is highly relevant, as the failure to restore proper myelin architecture is a central hallmark of progressive disability in human demyelinating diseases. [110]

Dopaminergic neurodegeneration

Traditional genetic models of Parkinson's Disease (PD) often exhibit weak or late-onset phenotypes, while neurotoxin-based models pose significant safety risks to researchers, limiting their scalability for high-content screening. To overcome these hurdles, Kim et al utilized a chemogenetic model in zebrafish to abate dopaminergic (DA) neurons. [113] NTR expression was driven from the *tyrosine hydroxylase* (*th*) promoter (the *th1* gene encodes an enzyme required for dopamine synthesis). *Th:NTR* fish express NTR1.0 in the DA neurons of the ventral forebrain, the zebrafish homolog of the mammalian substantia nigra. Exposure to MTZ in *th:NTR* fish caused substantial mitochondrial damage in DA neurons, characterized by mtDNA damage, dysfunction, diminished motility, and altered morphology, which ultimately resulted in DA neuron death. [113] This mechanistic parallel is highly relevant, as mitochondrial dysfunction is a central hallmark of human PD pathology. This finding elevates the model from a simple cell-killing assay to one of high pathological relevance, suggesting that the cellular stress induced by the NTR/MTZ system in this context faithfully recapitulates a fundamental aspect of the human disease. Consequently, the system provides a robust and scalable platform uniquely suited for the identification of therapeutic agents using smallmolecule screening.

NTR/MTZ-Based Screening

The NTR/MTZ ablation system provides a reproducible and scalable platform for functional screening in zebrafish, combining cell-type specificity, quantitative imaging, and compatibility with both chemical and genetic perturbations (Fig. 2). It enables systematic identification of compounds and genes that regulate injury response, cell survival, and regeneration across neural, hepatic, endocrine, and cardiac tissues.

Small-Molecule Screening

The NTR/MTZ ablation system has been adapted for high-content chemical screening in zebrafish, allowing quantitative evaluation of compound effects on cell death, protection, and regeneration across multiple tissues. In the context of retinal degeneration, [114] demonstrated the high-throughput capabilities of the system by screening 2,934 compounds using the *Tg(rho:YFP-NTR)* model of Retinitis Pigmentosa. By driving NTR specifically in rod photoreceptors, the Mumm lab induced targeted cell death and screened for small molecules that could preserve YFP-positive cells despite MTZ exposure. This large-scale effort identified 11 validated neuroprotectants, distinct from simple antioxidants, that were subsequently proven to show conserved efficacy in mouse retinal explant assays. [114] This cross-species validation confirms that the zebrafish NTR system effectively filters for compounds with relevant translational potential for human blindness.

Kim et al. (2022) utilized the *Tg(th:NTR)* model to perform a 1,403-compound screen for Parkinson's disease. By integrating automated imaging with rigorous statistical metrics including the Brain Health Score (BHS) and the Strictly Standardized Mean Difference (SSMD), the researchers identified 57 compounds that preserved dopaminergic neurons (Figs. 2A-B). Importantly, the study advanced beyond simple measurements of cell survival and provided mechanistic validation that these compounds protected neurons by restoring mitochondrial function, which is a central hallmark of PD pathology. The predictive validity of these hits was further confirmed through cross-assay validation in a separate Gaucher disease behavior model, demonstrating the system's capacity to identify robust therapeutics for complex neurodegenerative conditions (Fig. 2D). [113]

Another promising application of the NTR system lies in the identification of therapeutic agents that actively promote tissue regeneration. Lee et al. (2025) leveraged an optimized QF-based binary expression system (*mbpa:qf2;quas:epNTR-P2A-mCherry*) to perform a remyelination phenotypic screen for regenerative compounds. This transgenic line achieved greater than 85% oligodendrocyte loss following treatment with 2 mM MTZ for 18 hours, creating a highly reproducible regenerative baseline. Using this platform to screen a kinase-inhibitor library, the authors identified the TGF-β receptor I inhibitor AZ-12601011 as a potent driver of remyelination. [115] Mechanistic validation revealed that this compound promotes repair by modulating microglial and progenitor activation, thereby confirming the system's predictive validity for discovering clinically relevant restorative therapeutics that actively drive the reconstruction of functional tissue.

Similar regenerative screens have been successfully implemented in other tissues, such as the pancreas. Andersson et al. (2012) utilized the *Tg(ins:CFP-NTR)* line, crossed with a *Tg(ins:Kaede)* reporter to induce complete β-cell ablation and then monitor the formation of new β cells. This model was used in a high-content screen of approximately 7,000 small molecules to find compounds that would enhance regeneration of the insulin producing β cells. [116] This screen identified adenosine receptor agonists, specifically NECA, as potent stimulators of endocrine regeneration. Detailed mechanistic characterization revealed that NECA signals via the A2aa receptor to specifically enhance the proliferation of regenerating β-cells rather than neogenesis, a therapeutic pathway that was subsequently validated to restore normoglycemia in a streptozotocin-induced diabetic mouse model. [116]

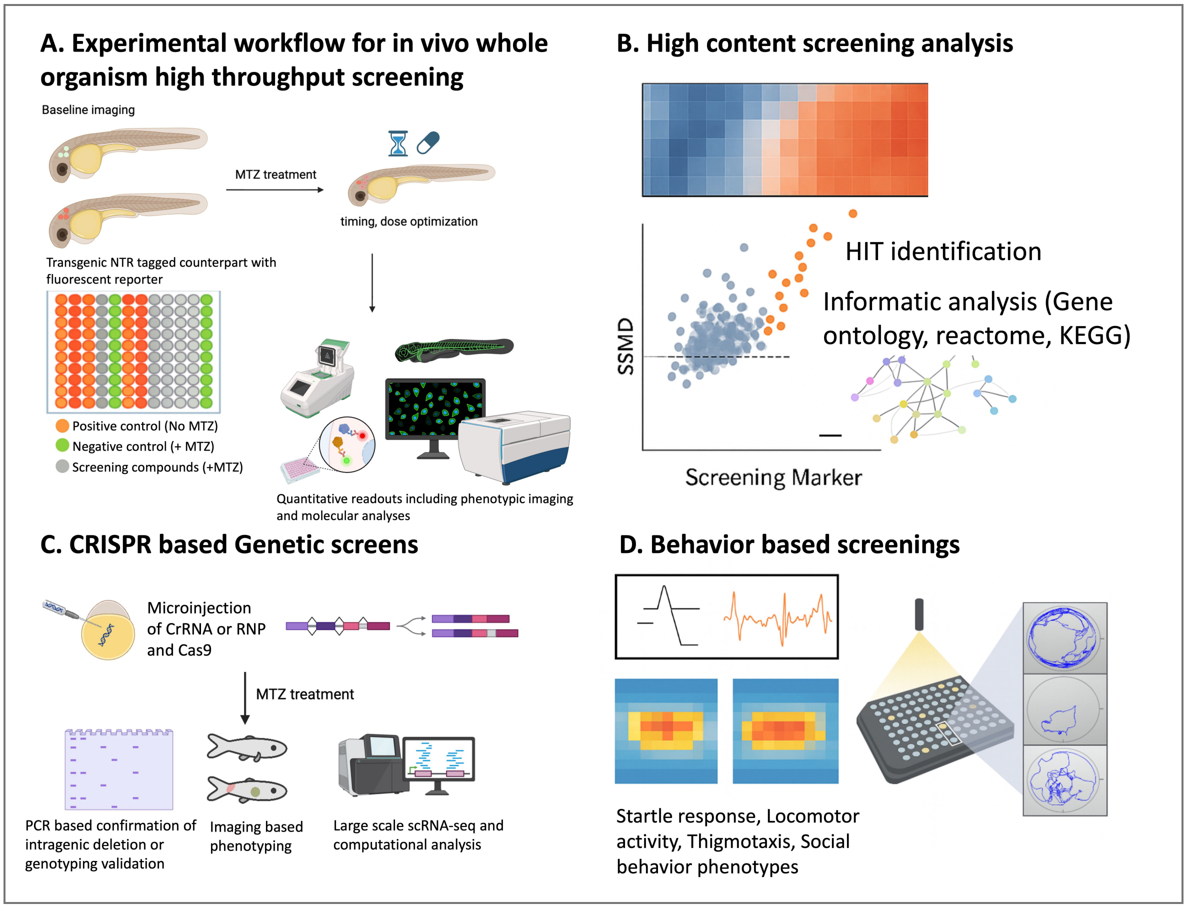

Figure 2. **NTR/MTZ-based screening platforms in zebrafish.**

Overview of the integrated chemogenetic screening workflow using nitroreductase (NTR)-mediated ablation. (A) Experimental design showing transgenic zebrafish expressing NTR in target tissues, baseline imaging, and subsequent metronidazole (MTZ) treatment to induce cell-type-specific ablation. The use of parallel transgenic controls and multiwell plate layout enables quantitative assessment of tissue loss and recovery. (B) High-content chemical screening pipeline integrating automated imaging, hit identification, and pathway-level analysis using standardized statistical metrics. (C) Genetic screening framework coupling sgRNA-based mutagenesis with imaging-based phenotype scoring to uncover modifiers of cell loss or regeneration. (D) Behavioral assays to quantify functional recovery or pharmacological response.

Genetic and CRISPR-Based Screening

Chemical screens can identify potential therapeutic reagents, though their molecular targets often remain unknown. A complementary approach is to perform reverse-genetic screens that integrate NTR-mediated ablation with CRISPR mutagenesis to identify genes affecting regeneration. [117] This mutagenesis is achieved by injecting Cas9 ribonucleoprotein (RNP) complexes multiplexed with several guide RNAs per target gene directly into NTR-transgenic embryos (Fig. 2C). This F0 'crispant' strategy generates high-efficiency somatic mutations in the first generation, [118] allowing researchers to induce cell-specific ablation with MTZ and immediately quantify the effect of gene disruption on regeneration without the delay of establishing stable mutant lines.

To identify regulators of Retinal Pigment Epithelium (RPE) repair, Lu et al. (2023) conducted a focused F0 CRISPR screen targeting 27 candidate genes in rpe65a:nfsB-eGFP larvae. [119] By injecting ribonucleoprotein (RNP) complexes containing three highly mutagenic guide RNAs per gene, they achieved high-efficiency somatic mutagenesis in F0 injected fish. The NTR/MTZ system induced the synchronized, widespread degeneration of the RPE, which subsequently triggered the secondary loss of photoreceptors. This screen identified numerous regulators of regeneration and revealed a novel mechanism that regulates the infiltration of phagocytic cells required for clearance of debris and complete regeneration. [119]

To find regulators of retinal ganglion cell (RGC) regeneration, Emmerich et al. (2024) performed a large-scale CRISPR screen onlOO genes. Using the isl2b:Gal4; UAS:YFP-NTR2.0 line for RGC ablation, they identified 18 effector genes comprising key transcription factors and signaling pathway components. [120] The screen revealed that inhibition of Asclla accelerated the regeneration of new RGC neurons.

Finally, the integration of F0 mutagenesis with automated imaging establishes a scalable framework for future genetic screens. The 'ZebraReg' platform utilizes a dual-transgenic line (tbx5a:CreERT2; myh7l:loxP-tagBFP-STOP-loxP-mCherry-NTR) that restricts NTR expression specifically to the heart ventricle. [100] Treatment with MTZ ablated approximately 97% of cardiomyocytes, triggering a robust regenerative response that typically restores the tissue within three days. By combining this precise injury model with F0 CRISPR mutagenesis followed by immediate phenotyping, the study demonstrates a proof-of-concept workflow to understand the genetic mechanisms of cardiac repair.

Caveats and improvements to the NTR/MTZ system

The NTR/MTZ system is widely used for diverse applications, but its performance has varied between labs. Key issues include batch-to-batch and preparation variability of MTZ, the need for high MTZ doses (≈10 mM) that can cause off-target toxicity (e.g., developing brain [121], larval/adult intestine [98], and differential susceptibility of some cell types to ablation [91]. Because ablation rate depends on both NTR activity and MTZ dose, researchers have pursued three complementary strategies to improve reproducibility and experimental interpretation: 1) increase NTR expression, 2) engineer higher-activity NTR mutants, and 3) identify more efficacious prodrugs that achieve effective killing at lower, less toxic concentrations. These iterative improvements are aimed at mitigating previous limitations and expanding the range of feasible experimental paradigms and are discussed next:

1) Increase NTR expression:

Strong, well-characterized promoters/enhancers (e.g., the zebrafish insulin promoter) [89] can drive high NTR expression, especially when present in multiple copies via Tol2-mediated transgenesis [122]. However, many cell-specific regulatory elements are weak, and maintaining multiple insertions is challenging and prone to genetic drift and intergenerational variability. An alternative is to use a bipartite system such as Gal4/UAS, [123,124] which can produce robust, amplified NTR expression even from single genomic insertion. With this approach, a cell-specific promoter drives a Gal4 transactivator that binds UAS sites to strongly activates *NTR* transcription (Figure 3). For example, elements from the 14xUAS constructs of Köster and Fraser [124] were used to generate the transgenic line Tg(UAS-E1B:NTR-mCherry)c264, [91,125]. These fish were distributed by the Zebrafish International Resource Center (ZIRC) and have been widely used by the zebrafish community: 10 of the 32 most-cited papers on zebrafish nitroreductase ablation use this line (S Table 1). A caveat is that Gal4/UAS DNA elements can be prone to epigenetic silencing, producing mosaic expression; the repetitive UAS contains multiple CpG sites susceptible to DNA methylation. [126,127] Silencing can be mitigated by using a less repetitive UAS (e.g., 4x) [128] or by using the QF/QUAS bipartite system (derived from Neurospora) [129], which has been reported to show reduced silencing [130]. and has recently been adapted for NTR-based ablation (Figure 3). [115,131].

2) Higher-activity NTR mutants:

Substantial effort has gone into engineering more active NTRs; first driven by their promise as cancer 'suicide-gene' therapies [132] and later to improve NTR-based ablation in basic research. [133] Two research groups independently engineered the same three substitutions into the wild-type *E. coli* enzyme (now termed NTR1.0), creating more efficient versions they named epNTR and NTR1.1.[132-134]. Cross-species screening identified a highly active nitroreductase (NTR) in *Vibrio vulnificus.* Using this enzyme as a scaffold, rational engineering yielded the second-generation variant NTR2.0, which exhibits a greater than 100-fold enhancement in activity over the original NTR1.0.135

The use of first-generation nitroreductase (NTR1) for chronic cell ablation was problematic, as the required 10 mM metronidazole (MTZ) dose induces intestinal pathology and approaches the LD50 in zebrafish. [98] However, the more active NTR2.0 variant enables effective ablation with far lower, better-tolerated MTZ concentrations. To demonstrate this, Tucker et al. developed a zebrafish model expressing NTR2.0 specifically in pancreatic β cells. [128] They found that efficient larval β cell ablation required only 100 μM MTZ, a regimen that could be maintained for 10 days without ill effects. In stark contrast, the NTR1 system required a toxic 10 mM MTZ dose, which is lethal to larvae (independent of NTR) within three days. In adult fish, a regimen of 5 mM MTZ for two days followed by two weeks at 1 mM was completely tolerated by wild-type fish with no ill effects but induced sustained hyperglycemia and weight loss in NTR2.0-expressing fish. This established a powerful model for studying chronic diabetic consequences, such as retinopathy, nephropathy, and impaired wound healing.

This well-tolerated ablation paradigm now makes it possible to model a range of other chronic conditions, including neurodegenerative, renal, and muscular disorders. This capability, in turn, facilitates the study of long-term disease progression and the evaluation of new therapeutic interventions.

3) More efficacious prodrugs:

Metronidazole (MTZ) efficacy can vary across suppliers and batches. To ensure consistency, it is recommended to prepare fresh MTZ solutions for experiments.[91,98] To overcome MTZ's limitations, alternative prodrugs like nifurpirinol (NFP) have been tested. NFP is a more potent nitrofuran-based prodrug.[91,98] However, its structural class is distinct from the nitroimidazole-based prodrugs for which NTR2.0 was specifically engineered.[135] As a nitroimidazole prodrug, Ronidazole (RNZ) likely retains compatibility with newer NTR systems while offering significant practical advantages over MTZ, primarily its better potency.[69,121,136] For instance, in Tg(fabp10:mCherry-NTR) fish, 2 mM RNZ achieved hepatocyte ablation comparable to 10 mM MTZ, a five-fold increase in potency.[136] This pattern of higher efficacy was replicated in a macrophage model, where a five-fold lower RNZ dose was as effective as MTZ.[121] Lai et al. also reported no bystander effects and demonstrated RNZ efficacy with the NTR1.1 variant. It has also been shown that RNZ functions with NTR2.0 to cause cell-specific ablation,[137] although a direct comparison of RNZ versus MTZ with NTR2.0 has not been reported. However, it is anticipated that the NTR2.0/RNZ combination will further lower the required prodrug concentrations and minimize off-target activity.

Given this evolving landscape of prodrugs and enzymes, what are the critical factors a researcher must weigh when designing an NTR ablation experiment?

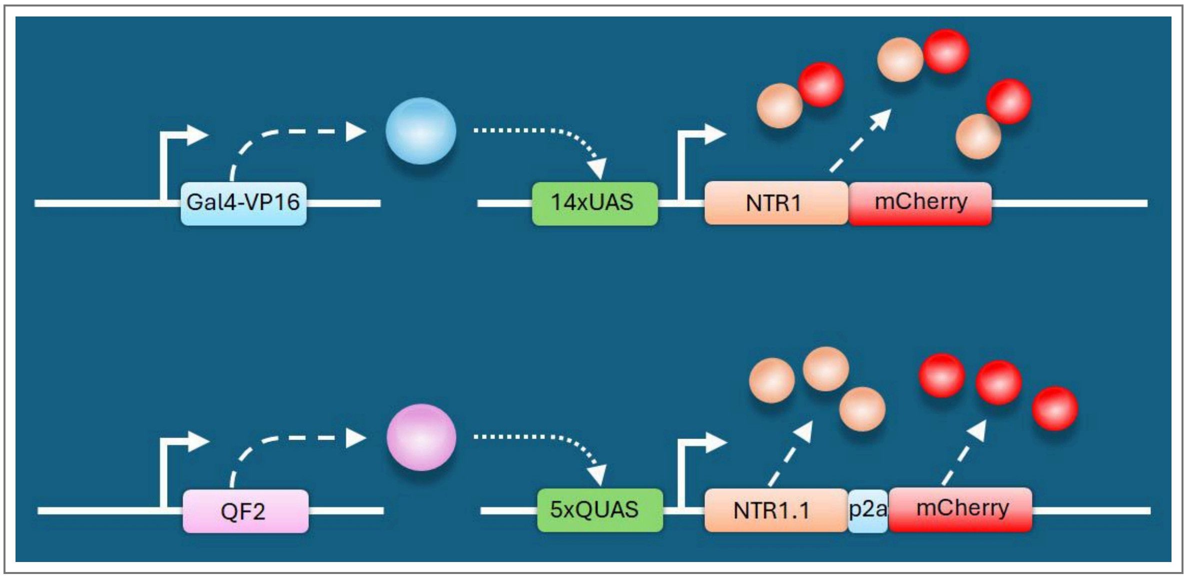

Figure 3. Schematic of bipartite systems to drive robust levels of NTR.

On left, driver lines lead to expression (dashed arrows) of transactivators (Gal4 or QF, blue, purple spheres) under the control of a regulatory element of interest (REI). These transactivators bind their upstream activating sequences (either UAS or QUAS, green boxes) to achieve controlled and amplified NTR expression (tan spheres) in target cells. NTR expression can be monitored by co-production of mCherry (red spheres) either as a fusion protein with NTR or as separate proteins due to P2A dependent ribosome 'skipping'.[139].

Experimental Design: Practical and Technical Considerations

A successful ablation experiment using the NTR/MTZ (or RNZ) system will require careful consideration of three key components: 1) transgenic strategy, 2) optimal NTR activity, and 3) appropriate controls. The specific biological question determines the optimal design. For regeneration studies, aim for complete ablation to easily assess for occurrence of neogenesis from progenitor cells. For functional studies, partial loss may be sufficient to observe a phenotype

1) Transgenic strategy:

Select regulatory elements carefully for tissue specificity; where a single promoter is insufficient, employ intersectional approaches (e.g., Cre/lox) to restrict NTR expression to the desired cell population. [138] Incorporate an independent fluorescent marker (fusion or 2A reporter[139]) to identify transgenic animals and to confirm cell-type specific expression.

Like any transgene, NTR transgenics may show positional effects such as leakiness and mosaicism, particularly with multicopy insertions. To ensure reliable lines:

– Screen multiple founders. Identify at least five independent F0 founders and ideally establish 5 F1 lines.
– Compare stable F1 lines to confirm that expression of fluorescent marker matches the published characterization of the transgene's regulatory elements.
– Prioritize a subset of F1 lines for further use based on the following criteria:

a. Mendelian transmission consistent with a single-site insertion, which simplifies downstream ablation experiments by avoiding variability from differing copy number.
b. Consistent expression: The linked fluorescent marker should show non-mosaic expression, confirming that NTR is expressed in all intended target cells. Validate by correlating the marker's fluorescence with independent methods like other reporter lines or antibody staining.
c. Robust, reproducible expression that is consistent regardless of whether the transgene is inherited from the male or the female.

2) NTR activity:

Often strong NTR expression produces faster, more complete ablation. [98,135] At the moment, NTR2.0 is the most active NTR used in zebrafish and there seems no reason not to use this particular enzyme in future ablation studies. If the promoter to be used to drive NTR2.0 is weak, consider amplifying expression via binary systems (Gal4/UAS or QF/QUAS).[115,125,131]

3) Controls:

To allow interpretation of ablation experiments, it is essential to validate that cell death was induced. Standard readouts include apoptosis assays, such as TUNEL and immunostaining for cleaved (active) caspase-3;[87,91] and monitoring of loss of fluorescent reporters.[89-91] If signal perdurance may confound interpretation, consider using destabilized reporters to minimize reporter longevity. [140,141]

A concern with NTR/MTZ ablation is that stressed target cells may downregulate both the NTR enzyme and its fluorescent reporter, allowing these cells to evade prodrug-induced death. While robust NTR expression and optimized dosing mitigate this risk, confirming the kinetics of cell death is often desirable. This can be achieved through endpoint analysis, *i.e.* fixing samples at serial time points and staining for apoptotic markers or by visualizing cell-death biosensors. HMGB1 is a nuclear chromatin-binding protein whose trafficking is death-mode specific: it is passively released during necrosis but exhibits strong nuclear retention during apoptosis.[142,143] By fusing HMGB1 to GFP, it is possible to create a transgenic biosensor that reports on cell death dynamics.[144] In zebrafish larvae co-expressing *ins:mCherry-2a-NTR2.0* and *ins:hmgb1-eGFP* in β-cells, [98,145] MTZ treatment triggered apoptosis marked by the loss of cytoplasmic mCherry signal and the concurrent unmasking of nuclei positive for HMGBl-eGFP (Fig. 4). A high dose of

MTZ (1 mM) induced apoptosis within 4 hours and complete loss of β-cell material by 24 hours, whereas a lower dose (10 μM) resulted in slower dynamics, with cellular debris still detectable at 24 hours. This approach demonstrates how cell-death reporters can provide feedback on timing and efficiency of cell death, information that is critical for optimizing prodrug regimens and interpreting ablation outcomes.

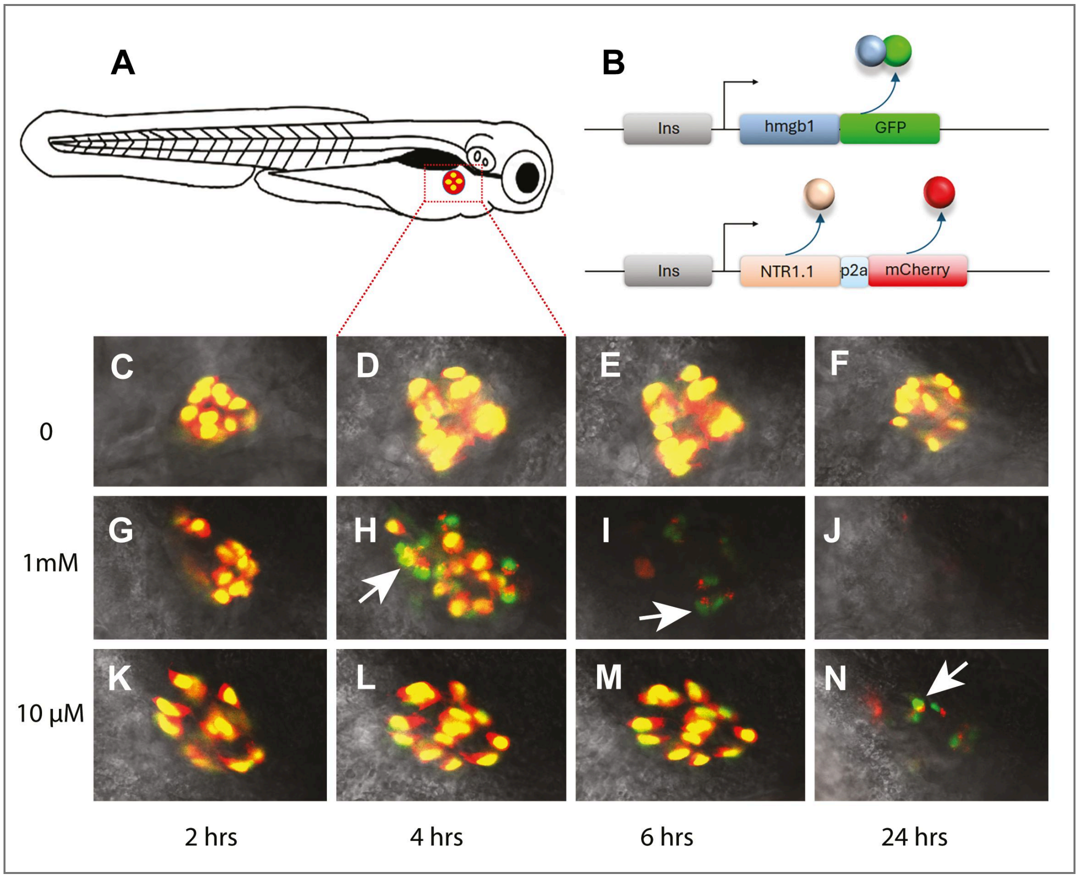

Figure 4. Live imaging of cell-death kinetics (A) Schematic of 6 dpf larvae showing region of fish imaged in C-N where the pancreatic islet is located (red/yellow). (**B**) Diagram of the two transgenes (*ins:Hmgb1-GFP*, *ins:mCherry-2a-NTR2.0*) in the fish in C-N. The insulin promoter (grey box) drives expression of the following: (**B**-above) an Hmgb1-GFP fusion protein and (**B**-below) NTR2.0 and mCherry [presence of the P2A (blue box) induces ribosomal skipping producing separate proteins]. **C-N** Confocal images of the islet of in three larval fish over a time course from 6 dpf to 7 dpf (times along the X axis). **C-F** negative control – no MTZ (0). **G-J** fish treated with high MTZ dose (1mM). **K-N** fish treated with a low dose MTZ (10μM). **G-N** Dying β cells first lose red fluorescence, revealing green nuclei (arrow heads). A higher dose shows appearance of green nuclei (**H**) earlier than the lower dose (**N**). **J** 24 hrs in 1mM and no debris remains.

Finally, in any NTR-based ablation experiment there should be two negative controls:

- NTR transgene, no prodrug - controls for effects of exogenous NTR expression and provides base line response.
- No NTR transgene, prodrug - controls for off-target prodrug effects, including antimicrobial activity (nitroaromatic prodrugs such as MTZ/RNZ will affect the microbiome). For host–microbiome studies, consider alternative ablation methods.

These guidelines are intended to help researchers design robust, interpretable NTR-based ablation experiments, facilitating investigations into important biological questions.

Additional files

Supplemental table 1

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Peer reviews

Reviewer #1 (Public review):

Summary:

Kim and Parsons present a timely overview of the NTR/prodrug system and its applications in regenerative biology research, with particular emphasis on tissue-specific cell ablation. The system has substantially advanced the field by enabling non-invasive, conditional cell elimination, and has proven especially powerful in zebrafish, though applications in other classical model organisms are also noted. The review covers the historical origins of the NTR system, its use in regeneration studies, small-molecule screening, and genetic and CRISPR-based screening, as well as future directions, including the development of the highly efficient NTR2 enzyme variant.

Strengths:

This is a useful and well-structured contribution. The manuscript is a valuable resource for the regeneration biology community.

Weaknesses:

The impact and scientific value of this paper could be meaningfully enhanced by addressing several points outlined below. The concerns centre on completeness, conceptual precision, and the depth of mechanistic discussion.

(1) Title: Species specificity.

Given that the review's primary focus is the zebrafish model, it would be appropriate to include the species name in the title. This would improve discoverability and accurately set the scope of the article for prospective readers.

(2) Subchapter: Physical injury.

The subchapter enumerates different types of physical injury models but would benefit from a more substantive comparative discussion. In particular, the authors are encouraged to address the following:

(2.1) Outcome comparison: Surgical and other invasive approaches cause damage to entire tissue structures comprising multiple cell types, whereas tissue-specific genetic ablation eliminates a defined cell population while leaving the surrounding architecture largely intact. This fundamental distinction has direct implications for the interpretation of regenerative outcomes and should be clearly articulated.

(2.2) Inflammatory response: Invasive injuries typically trigger a robust inflammatory response, which itself can be a potent driver of regeneration. By contrast, genetic cell ablation may elicit a qualitatively different inflammatory reaction. A comparative discussion of this distinction would help readers appreciate a critical limitation of genetic ablation systems relative to models of natural, accidental tissue damage.

(3) Subchapter: Cell-specific toxins.

This subchapter would benefit from several targeted expansions:

(3.1) Off-target effects: The authors should include evidence that the exemplified drugs have known off-target activities, with a discussion of how these confounded the interpretation of experimental data. At least a few concrete published examples should be cited.

(3.2) Completeness of the toxin list: The current list appears illustrative rather than comprehensive. A more complete enumeration would be valuable, particularly for neurotoxins and drugs targeting sensory cells, as these are highly relevant to the zebrafish regeneration field.

(3.3) Interspecies differences: It would be informative to specify whether drug specificity differs across species, as this is a practical consideration for researchers working in organisms other than zebrafish.

(4) Subchapter: Optogenetic cell ablation.

The authors note that optogenetic cell ablation has not yet been applied in conventional regeneration studies. It would strengthen this section to include a discussion of the underlying reasons for this gap, whether technical or biological, so that readers can appreciate the barriers and potential for future adoption.

(5) Terminology: "Suicide gene".

The use of the term "suicide gene" to nitroreductase is conceptually imprecise and merits reconsideration. Strictly speaking, a suicide gene is one whose expression alone is sufficient to kill the cell, as in the case of genes encoding direct triggers of apoptosis or the catalytic A subunit of diphtheria toxin (DTA). NTR does not meet this criterion: it requires the exogenous administration of a prodrug (e.g., metronidazole) to produce a cytotoxic metabolite, and is therefore only conditionally lethal.

It is worth noting that nitroreductases evolved in bacteria and fungi as enzymes involved in chemoprotection and detoxification, converting potentially toxic and mutagenic nitroaromatic compounds into less harmful metabolites (PMID: 18355273). This biological context further underscores that NTR is not inherently a lethal protein. The authors are encouraged to replace or qualify the term "suicide gene" and instead adopt terminology that more accurately reflects the conditional, prodrug-dependent nature of the system.

(6) NTR/MTZ in regenerative studies: Mechanistic depth.

While the review catalogues several studies employing the NTR/MTZ system, it lacks mechanistic depth regarding the cellular basis of ablation. The following questions should be addressed, where evidence exists in the literature:

(6.1) Temporal dynamics of cell death: What is known about the kinetics of NTR/MTZ-induced lethality across different tissue types in larval and adult zebrafish, as well as other organisms? Are there age- and tissue-specific differences in the speed or completeness of ablation?

(6.2) Mechanism of cell death: What is the cellular basis of NTR/MTZ-induced cytotoxicity in zebrafish? In particular, do the toxic metabolites preferentially cause mitochondrial damage or nuclear DNA damage, and what downstream death pathways are engaged?

(6.3) Proliferative versus post-mitotic cells: Are proliferating and non-proliferating cells equally sensitive to the NTR/MTZ system, or does the proliferative status of a cell influence susceptibility? This is a practically important question for researchers designing ablation experiments in tissues with mixed cell populations.

(6.4) Ablation of progenitor cells: Are there published examples demonstrating that co-ablation of differentiated functional cells and organ-specific progenitor cells abolishes regenerative capacity? Such examples would be highly informative in illustrating the system's power to dissect the cellular requirements for regeneration.

Addressing the points above, particularly the comparative discussion of injury models and inflammatory responses, the clarification of terminology, and the mechanistic discussion of NTR/MTZ-induced cell death would substantially strengthen the review's scientific contribution and utility.

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Reviewer #2 (Public review):

Summary:

Kim and Parsons reviewed the nitroreductase (NTR)/prodrug system: when engineered cells expressing the enzyme NTR are treated with prodrug (e.g. metronidazole), NTR converts the prodrug into a cytotoxic compound that kills these cells. The review covers how the system has been developed, spatiotemporal control of targeted cell ablation, and its broad utility to study regenerative mechanisms, model human diseases, and screen chemicals to discover pro-regenerative and protective compounds. They further discussed the newer version of NTR, a more potent prodrug, and experimental design, which not only expands the possible utility of the NTR/prodrug system, but also allows the research community to develop a precise, reproducible and versatile platform.

Strengths:

The review summarized landmark work application of the NTR/prodrug system, and recent studies, with focus on the model organism zebrafish. The review provides a good gateway to understanding the system and considering regenerative studies.

Weaknesses:

No weaknesses were identified by this reviewer.

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Reviewer #3 (Public review):

Summary:

This manuscript by Kim and Parsons presents an overview of the nitroreductase/metronidazole (NTR/MTZ) cell ablation system.

Strengths:

This manuscript nicely places the NTR/MTZ system in the context of other cell ablation methods, with a discussion of their respective advantages and disadvantages. This review is particularly useful for highlighting the many ways the NTR/MTZ system has been applied to study the regeneration of multiple cell types and to model different degenerative human diseases. The review concludes with a discussion on recent improvements made to the system and practical considerations and "best practices" for NTR-based experiments. This review could be a helpful resource, especially for researchers new to regeneration or cell ablation studies.

Weaknesses:

Although the NTR/MTZ system has been used in other model organisms, this review is primarily focused on its uses in zebrafish. While this is understandable given the wide adoption of NTR/MTZ in the zebrafish field, discussion of the unique considerations and/or challenges for non-zebrafish systems would be an interesting addition and could broaden the potential audience for this review. Additional minor revisions, as suggested below, could also improve readability.

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